

6th June, 2026

AI-POWERED TABLET VISUAL INSPECTION SYSTEM FOR PHARMACEUTICAL MANUFACTURING

Milestone 3 - Model Evaluation and
Training

Presented by:

Sowmya

Lalitha Saranya

Deepak Reddy

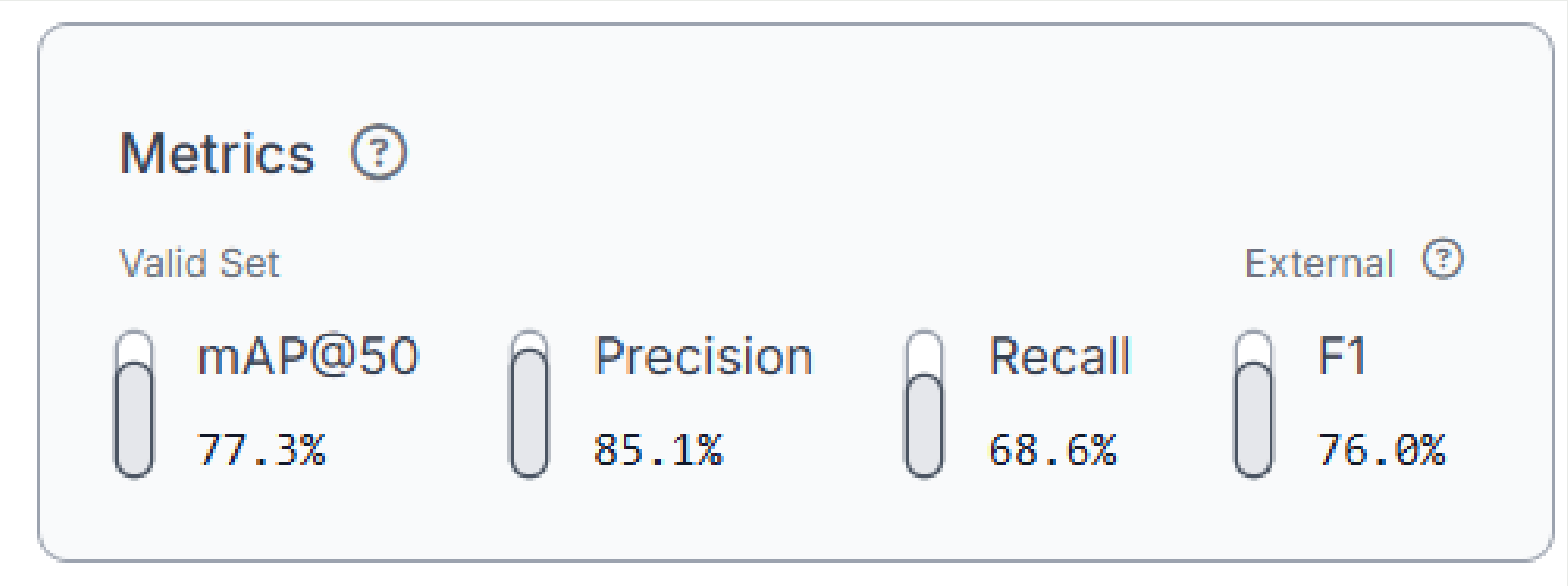
Naga sriya santoshi

MODEL-1: YOLOV11 OBJECT DETECTION (NANO)

Model Specifications:

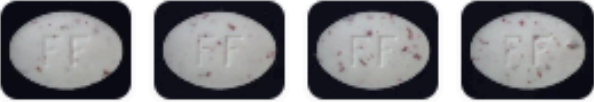
Specification 	Details
Model Name	YOLO11n (Nano)
Input Size	640x640 pixels
Parameter Count	~ 2.6 Million
mAP ₅₀₋₉₅ (on COCO)	~ 39.5%
Inference Speed (CPU)	~ 56 ms
Inference Speed (Edge/GPU)	~ 1.2 - 3 ms

Performance Specifications



SAMPLE TEST IMAGE - FAULTY PRINT

Test Set



[View Test Set →](#)

Drop Files or

+ Select Files

Image or Youtube URL...

Try with:

My Webcam

My Machine



1 object detected

Model Visualizations

Confidence Threshold: 50%

0% 100%

Overlap Threshold: 50%

0% 100%

Opacity Threshold: 75%

0% 100%

Label Display:

Draw Labels

Show Ground Truth

Output

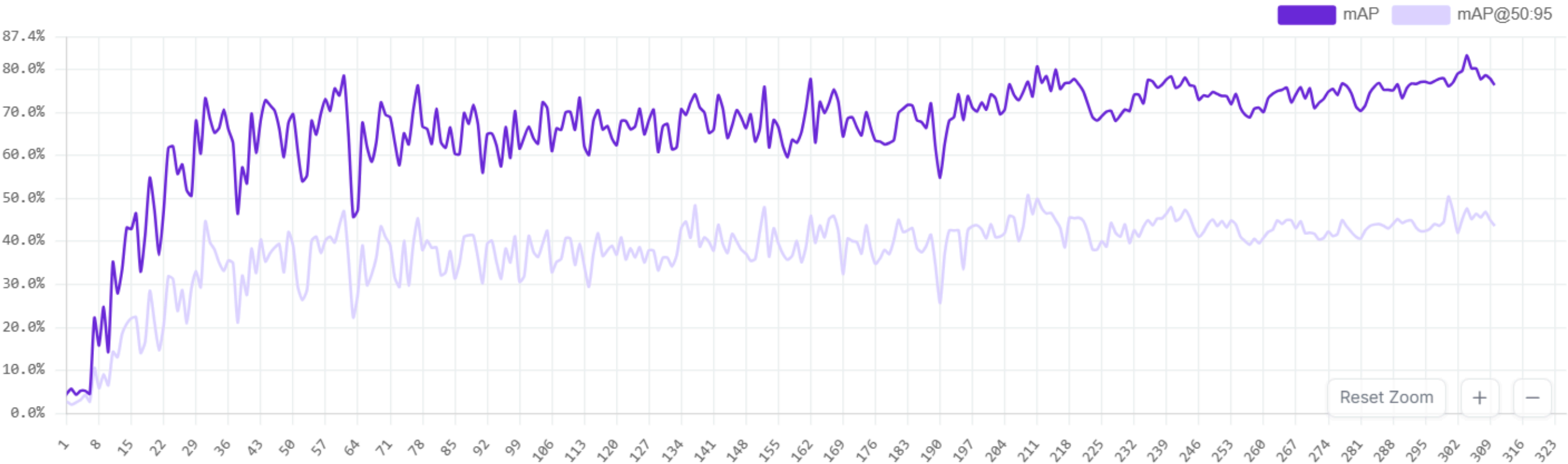
```
{  "predictions": [    {      "x": 246,      "y": 249.5,      "width": 162,      "height": 101,      "confidence": 0.878,      "class": "faulty pri      "class_id": 3,
```

Training graphs

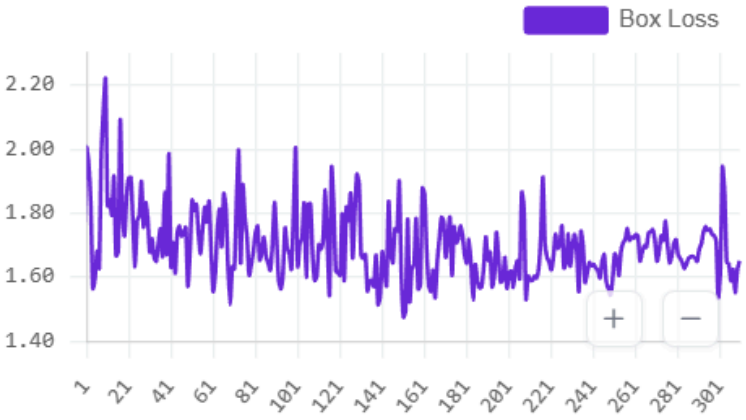
Training Graphs

Advanced Graphs

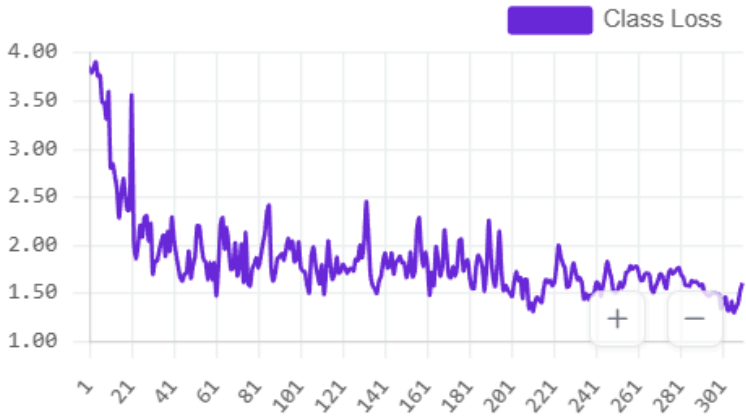
Model Performance



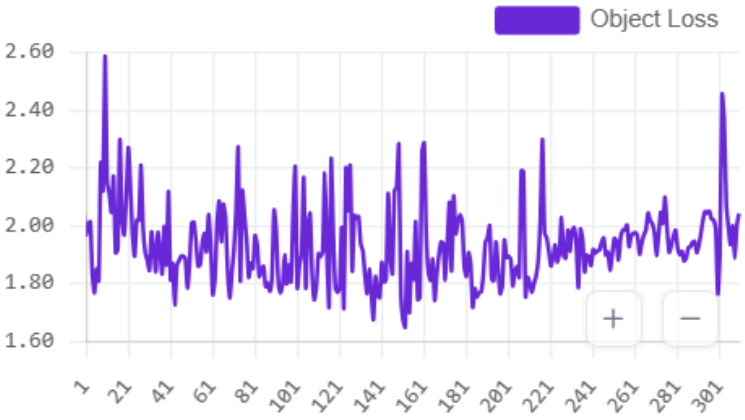
Box Loss



Class Loss



Object Loss

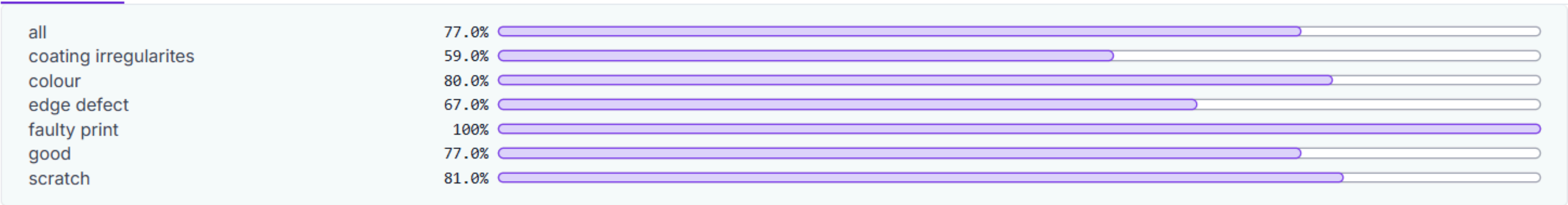


Training graphs

Average Precision by Class (mAP50)

These metrics are reported by the model provider and may not follow industry-standard evaluation techniques. Learn more about model evaluation best practices and important caveats [here](#).

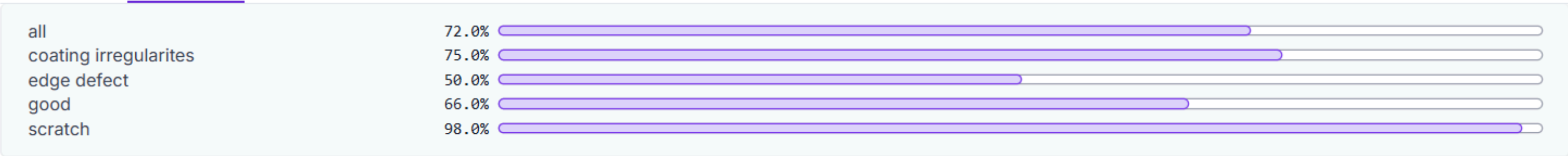
Validation Set Test Set



Average Precision by Class (mAP50)

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Validation Set Test Set



MODEL-2: ROBOFLOW RF-DETR

Model Specifications:

Architecture Family: Detection Transformer (DETR)

Native Input Resolution: 880× 880 pixels

Task Capabilities: Designed for high-precision Object Detection

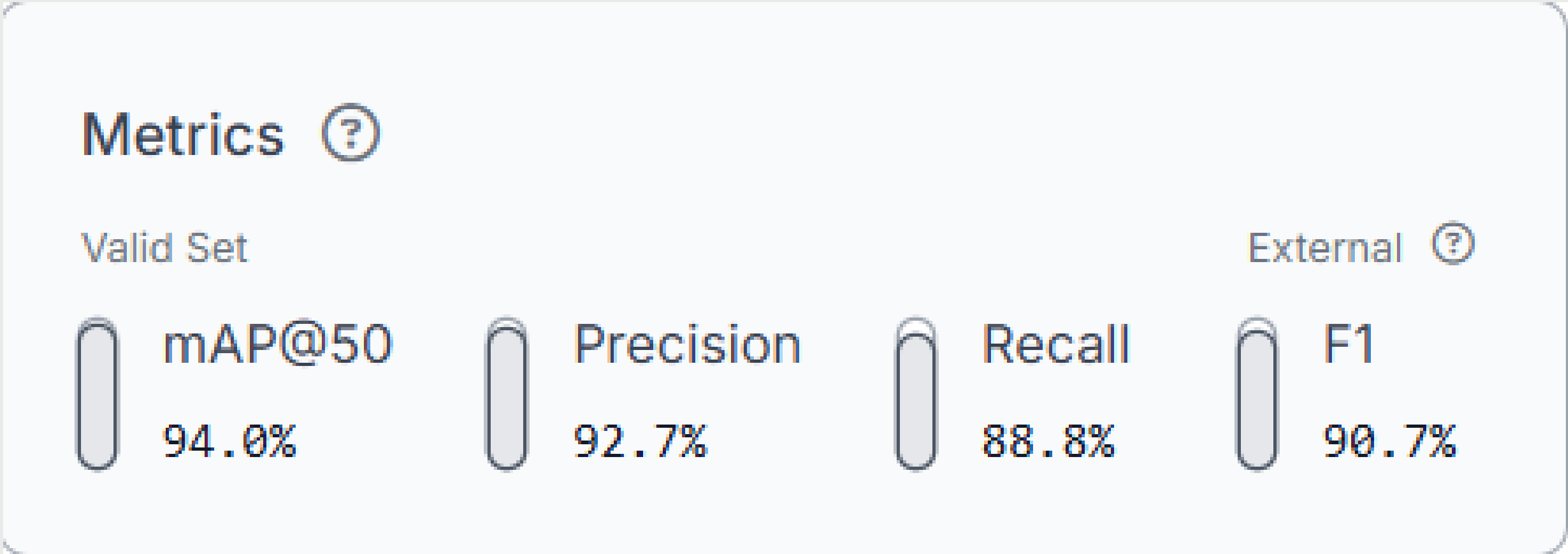
Inference Speed: ~17.2 ms per frame (when tested on an NVIDIA T4 GPU using TensorRT FP16)

Preprocessing Used: Auto-Orient

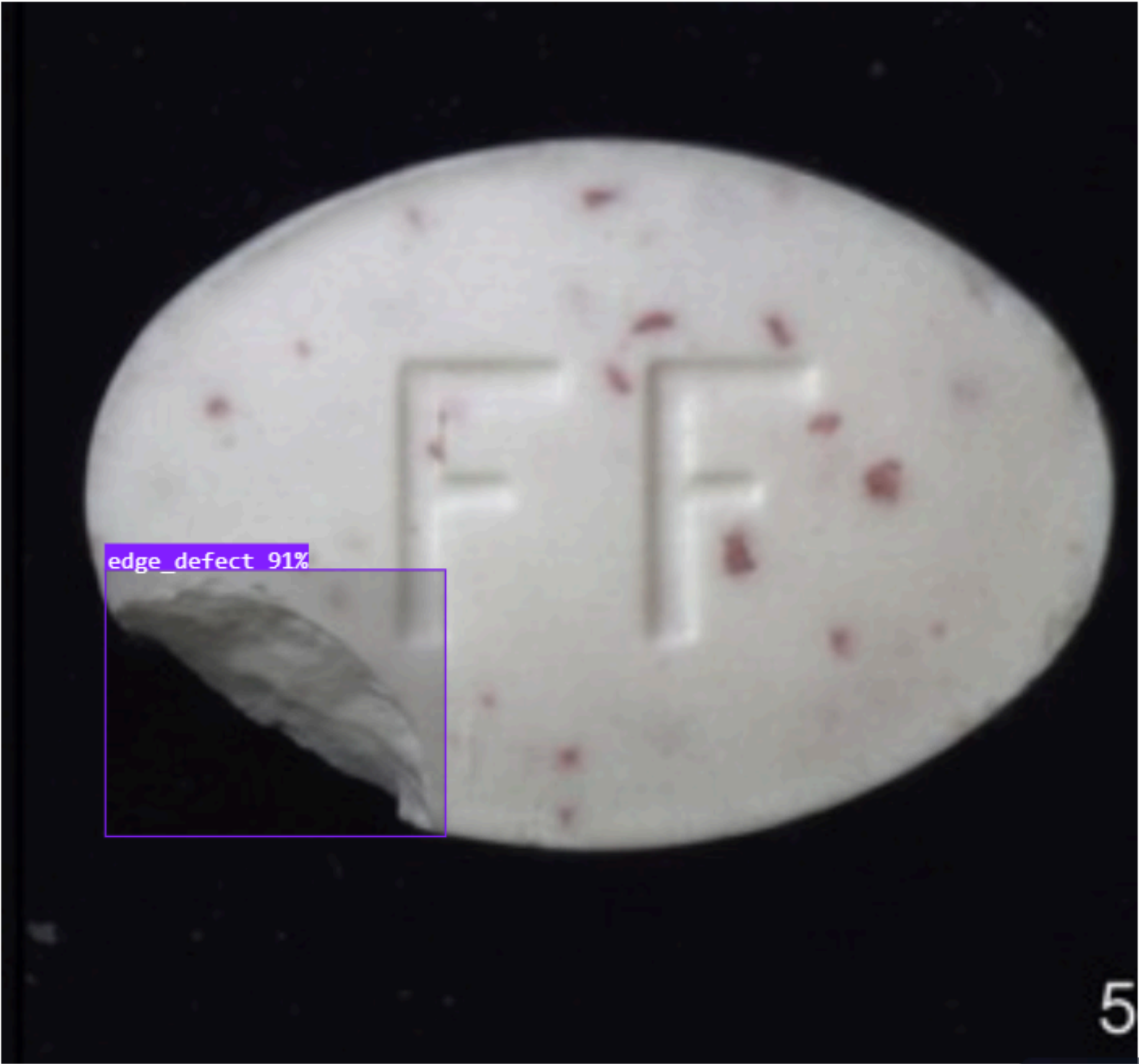
Augmentations Used: Flip (Vertical and Horizontal), Blur (-10% to +10%), Exposure (-10% to +10%), 90 degree rotate

Total Images: 428 (Train-354, Valid-50, Test-24)

Performance Specifications



SAMPLE TEST IMAGE - EDGE DEFECT



edge_defect 91%

5

1 object detected

Model Visualizations

Confidence Threshold: 50%

0% 100%

Opacity Threshold: 75%

0% 100%

Label Display:

Draw Confidence

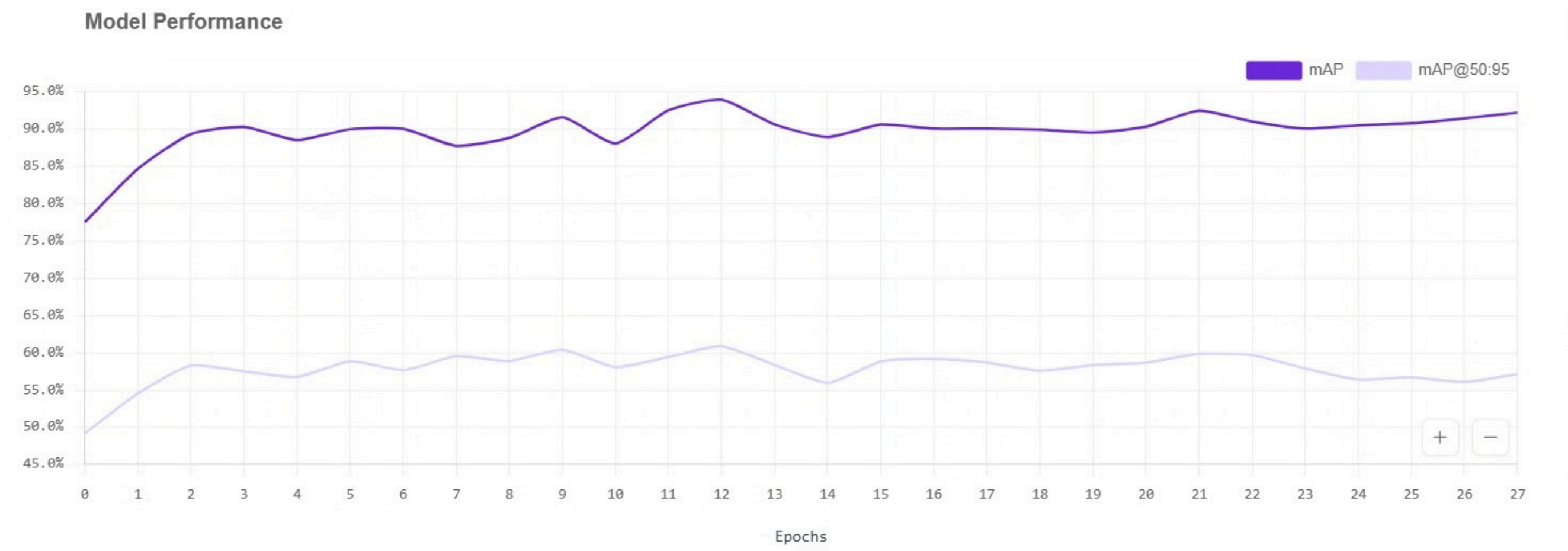
☒ Show Ground Truth

Output

```
predictions": [  
  {  
    "x": 71,  
    "y": 181.5,  
    "width": 88,  
    "height": 69,  
    "confidence": 0.911,  
    "class": "edge_defect",  
    "class_id": 2,  
    "detection_id": "eae59142"  
  }  
]
```

Training graphs

Training Graphs

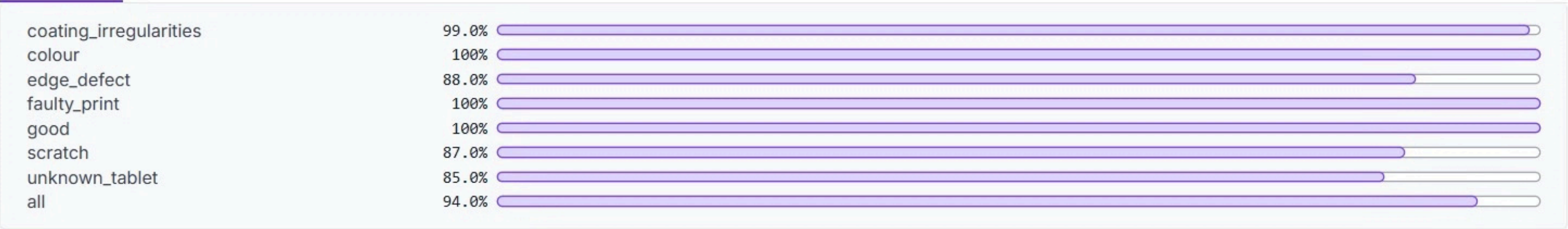


Training graphs

Average Precision by Class (mAP50)

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Validation Set Test Set



Average Precision by Class (mAP50)

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Validation Set **Test Set**



THANK YOU